

SEA-LION powered Language Support

- English
- Chinese
- Indonesian
- Thai
- Vietnamese
- Filipino Tagalog
- Malay
- Burmese

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SMART Image Medical Instruction System



The **Silent Crisis** in Home Healthcare

The Problem: Incorrect use of medical devices is a major, often overlooked, issue. It leads to misdiagnosis, delayed treatment, and can cause over 1,000 preventable deaths annually in Southeast Asia alone. Language barriers and complex instructions make it even more difficult for the 650 million people in the region who manage chronic illnesses at home.

The High Stakes: A single mistake with a blood pressure monitor, thermometer, or glucose meter can invalidate readings, causing life-threatening emergencies.

Introducing **SIMIS**: Your AI Medical Assistant

Our Solution: SIMIS (Smart Image Medical Instruction System) is a production-grade medical device assistant that uses AI and computer vision to provide real-time guidance. It's designed to make using medical devices safer, easier, and accessible to everyone.

The Goal: To prevent medical errors, improve health outcomes, and save lives.

MDR Data Files

Medical Device Reporting (MDR): How to Report Medical Device Problems

[Exemptions, Variances,
and Alternative Forms of
Adverse Event Reporting
for Medical Devices](#)

[Voluntary Malfunction
Summary Reporting
Program](#)

Each year, the FDA receives over two million medical device reports of suspected device-associated deaths, serious injuries, and malfunctions. These medical device reports have been available in the [FDA's public Manufacturer and User Facility Device Experience \(MAUDE\) Database](#) since 1999 as one source of information to help patients and health care providers make more informed medical decisions.

On this page, the FDA is also providing the medical device reports received from 1984 - 1996 through CDRH's legacy Device Experience Network (DEN) reporting system and reports received under the Alternative Summary Reporting Program from 1999 – April 2019.

Under the Freedom of Information and Privacy Acts (Sec. 552, Title 5, USC) (PL 93-579), the FDA is required to delete, prior to public disclosure, any information that constitutes trade secrets, and confidential, commercial, or financial information; and any personal, medical, and similar information that would constitute a clearly unwarranted invasion of personal privacy. Included in the deletion requirements are all identification of the reporters of the events and the user facility where the event occurred.

Content current as of:
09/09/2025

Regulated Product(s)
Medical Devices

The Silent Crisis in Home Healthcare

- FDA reports 2 million+ device adverse events annually

MDR Data Files

**Medical Device
Reporting (MDR): How to
Report Medical Device
Problems**

[Exemptions, Variances,
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Regulated Product(s)
Medical Devices

The **Silent Crisis** in Home Healthcare

- FDA reports 2 million+ device adverse events annually
- 28% involve user interaction issues

Article

An Analysis of FDA Adverse Event Reporting Data for Trends in Medical Device Use Error

September 2020 · Proceedings of the International Symposium on Human Factors and Ergonomics in Health Care 9(1):130-134

DOI:[10.1177/2327857920091024](https://doi.org/10.1177/2327857920091024)

The Silent Crisis in Home Healthcare

- FDA reports 2 million+ device adverse events annually
- 28% involve user interaction issues
- ASEAN's aging population (77 to 173 million by 2050) needs safer home healthcare

Region/Sub-region	2020		2050	
	N ₆₀₊ ('000)	% ₆₀₊	N ₆₀₊ ('000)	% ₆₀₊
World	1,060,593	13.5	2,132,389	22.0
Africa	73,724	5.4	215,0686	8.7
Asia	619,139	13.3	1,337,587	25.3
Western Asia	23,4394	8.2	76,074	19.0
Central Asia	6,620	8.9	16,296	15.7
Southern Asia	189,219	9.6	474,541	19.2
Eastern Asia	322,478	19.4	597,330	39.2
South-Eastern Asia	77,383	11.5	173,346	22.0
Europe	191,351	25.6	252,054	35.9
Latin America & the Caribbean	83,799	12.9	188,111	25.14
Northern America	85,091	22.8	125,644	29.8
Oceania	7,489	17.0	13,925	24.1

Source: United Nations, 2022 (World Population Prospects 2022)



Camera input



Custom YOLOv8
models



SEA-LION



Real-time output,
culturally-aware
feedback





Camera input

Custom YOLOv8
models

SEA-LION

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What device do you need help with?

Select Your Device



Oral Thermometer

Check your temperature orally



Ear Thermometer

Check your temperature via ear

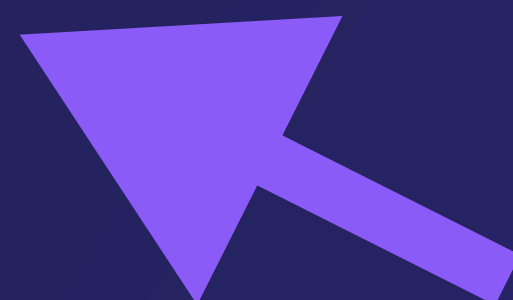


Blood Pressure Monitor

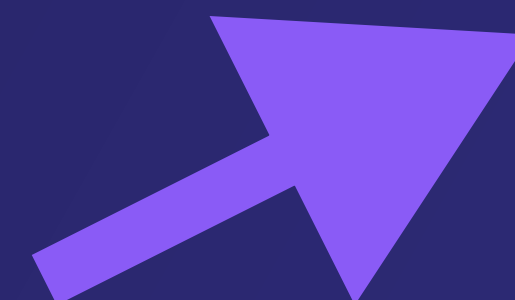
Monitor your blood pressure



Our Team



Pharmacists





Our Team



AI Engineer



Our Team



AI Believers



Our Team



Asia Pacific home healthcare

Asia-Pacific Home Healthcare Market

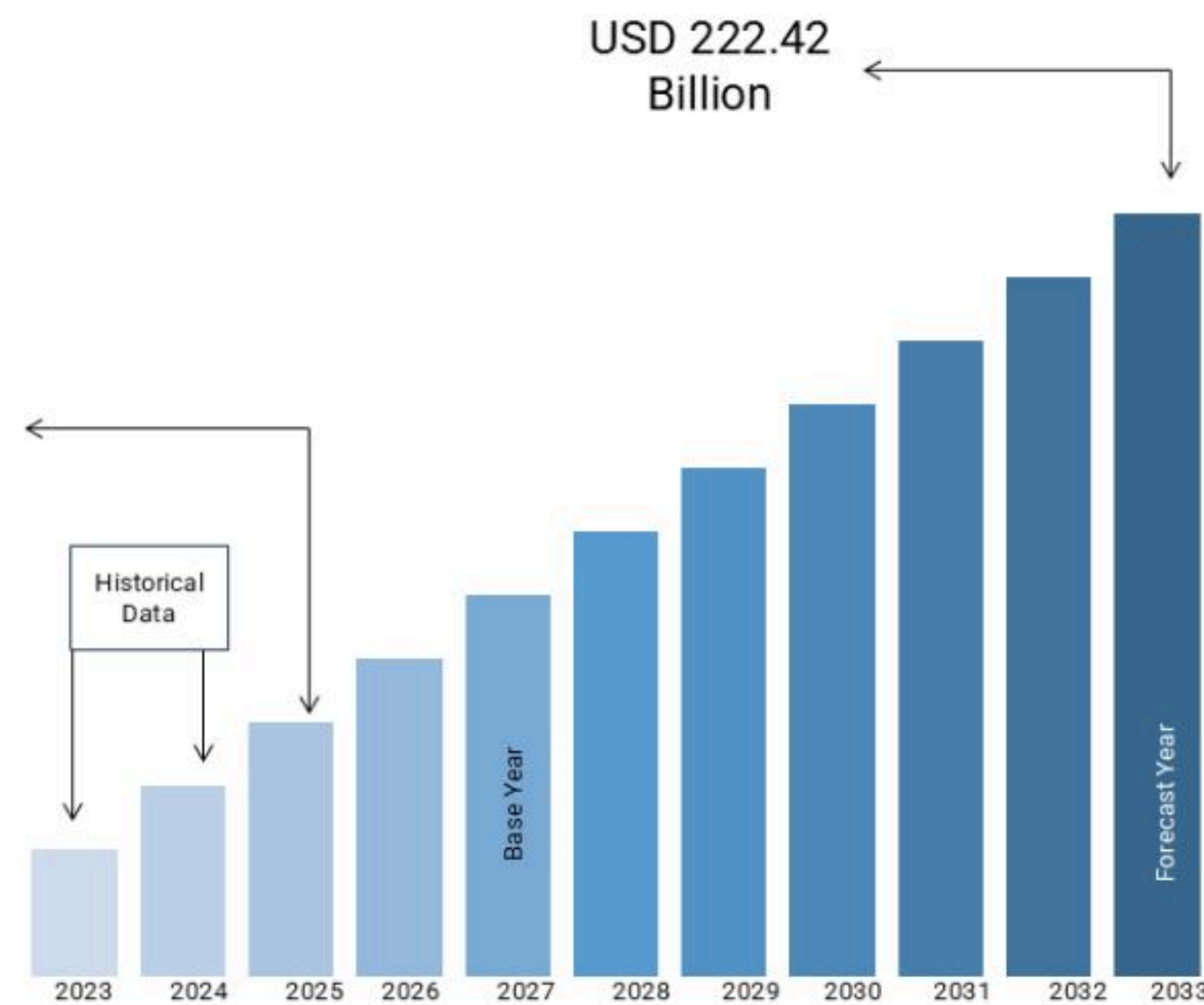
Market Size Overview



11.48%

Europe market CAGR,
2025 - 2033

USD
93.24
Billion





Smart devices, dumb instructions — SIMIS fixes that for 650 M Southeast Asians



The **Silent Crisis** in Home Healthcare

- 28% of device errors involve user issues (WHO 2023)
- 10 languages create instruction barriers
- \$36.3B home healthcare market needs safer guidance



Performance Innovation

- Intelligent database caching of 105 pre-generated guidance entries
- 25–131 ms response times—95% faster than pure AI generation
- 99.8% cost reduction—from \$0.50 to \$0.001 per request

Production-Grade Infrastructure

Fully serverless AWS infrastructure
Lambda, API Gateway
99.9% uptime

Designed for SEA environment

Accessible even on entry-level devices
Cloud and edge computing
Loads fast even on slow rural connections



Demo Video



222 Billion Market

We are the first AI-driven situational learning platform in Southeast Asia that combines medical accuracy, multilingual support, and a unique P3 (Prepare, Practice, Perform) framework. This focus on real-world skills is our key competitive advantage.



SEA Talent Crisis

Region faces a talent crisis where graduates struggle to apply knowledge in real-world scenarios



Solving Real-World Healthcare Needs Today

SIMIS is the first AI-driven situational learning platform in Southeast Asia that combines medical accuracy, multilingual support, and a unique P3 (Prepare, Practice, Perform) framework.

Growing Device Support

SIMIS provides comprehensive guidance for digital thermometers and blood pressure monitors. We are actively developing guidance for blood glucose meters, digital scales, and pulse oximeters. We are continuously expanding our coverage to support a growing library of essential home healthcare needs.

Wide multilingual use

Breaking down communications barriers, our platform supports English, Chinese, Indonesian, Thai, Vietnamese, Filipino, Malay, and Burmese. This ensures clear, culturally-accurate guidance, making complex medical instructions accessible to the unique diverse communities across Southeast Asia.

Fast accessible guidance

SIMIS offers unparalleled ease of use. Users can access our AI assistant instantly from any web browser on any device, with no need for account registration or app installation. This frictionless, zero-barrier approach makes critical health information readily available, even in rural areas and the growing silver generation.



Every Day of Delay Means More Preventable Deaths

- Our Ask: We are seeking support to scale our life-saving AI guidance across SEA and beyond
- Our Vision: To transform home healthcare and make medical technology accessible to everyone,
- everywhere.

Join Us in Revolutionising Healthcare

Let's build a bridge between complex medical technology and everyday users, one life at a time.

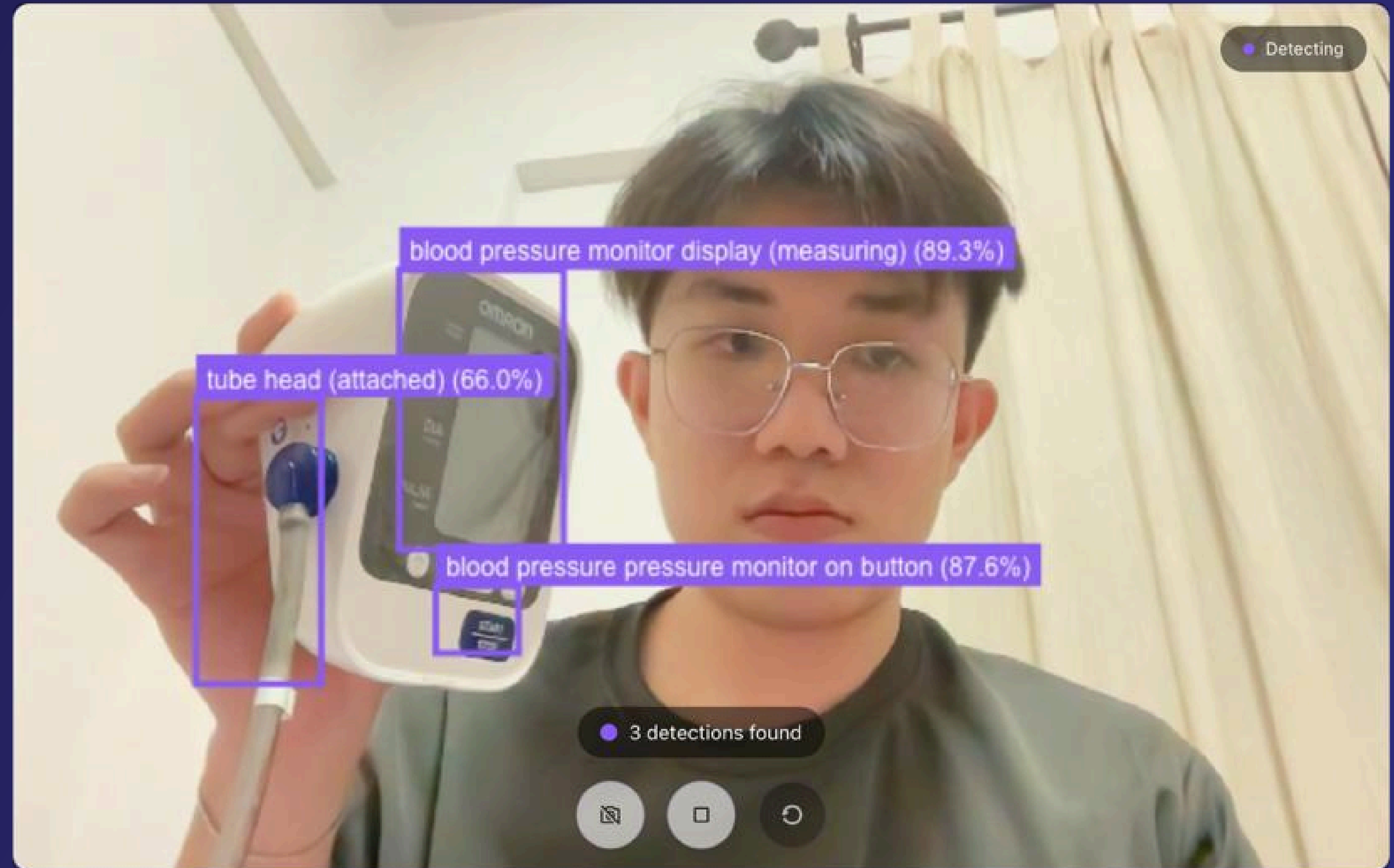


Camera input

Custom YOLOv8
models

SEA-LION

Real-time output,
culturally-aware
feedback



First healthcare application built specifically for ASEAN contexts



Why SEA-LION Matters in Healthcare

- Generic LLMs lack ASEAN medical context understanding
- SEA-LION provides culturally-appropriate healthcare guidance in 7+ regional languages
- Reduces instruction errors by using native language in text and audio
- Ensure culturally-aware guidance, especially for the silver generation

Quần băng tay cao 2 cm so với khuỷu tay

Bungkus manset 2 cm di atas lipatan siku

Balutin ang manseta 2 cm sa itaas ng siko

พันเฟือกให้แน่น 2 ซม. เหนือข้อพับแขน



Balut gelang 2 cm di atas lipatan lengan

រុំក្រវ៉ាត់ ២ សង់ទីម៉ែត្រលើសព័កូនដៃ

ພັນແຂນໃຫ້ແໜ້ນ 2 ຊມ ເໜືອຂໍແຂນ

ဘယ်လက်ဖျားထက် ၂ စင်တီမီတာ အထက်မှာ ချုပ်ပါ



The **Silent Crisis** in Home Healthcare

- FDA reports 2 million+ device adverse events annually
- 28% involve user interaction issues
- ASEAN's aging population (77 to 173 million by 2050) needs safer home healthcare



High Stakes

A single mistake with a blood pressure monitor, thermometer, or glucose meter can lead to **life-threatening emergencies**



How **SIMIS** Works

Smart Device Detection

Custom-trained YOLOv8 computer vision identifies medical devices, preventing wrong device usage

Multilingual Guidance

Step-by-step text/audio instructions in 7+ languages with culturally adapted medical terminology

24/7 AI Assistant

Real-time chat assistant available around the clock for troubleshooting and expert guidance

Progress Tracking

System monitors progress for each step and provides visual confirmation to ensure compliance

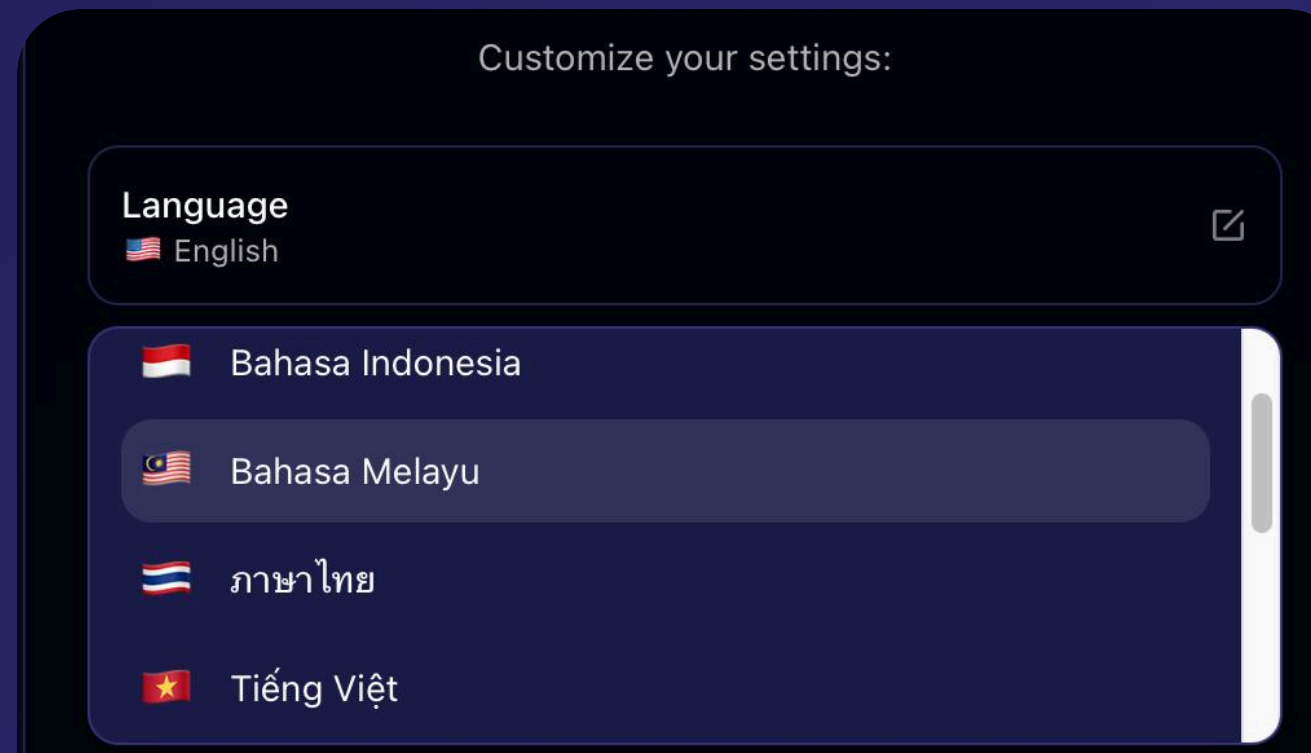


Solving Real-World Healthcare Needs Today



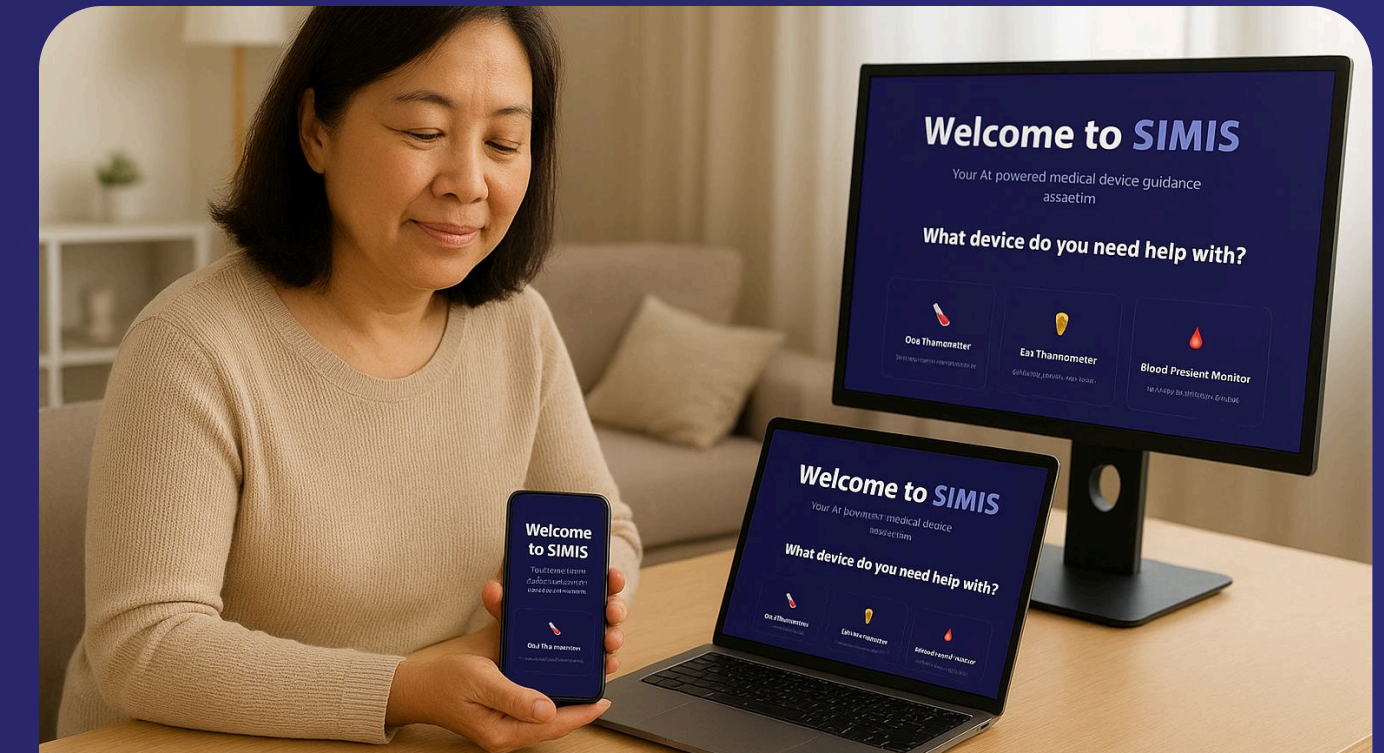
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SIMIS Roadmap

Phase 1 (0-6 Months): Market Penetration

Expand medical device coverage and establish partnerships with telemedicine platforms and hospitals.

Phase 2 (6-12 Months): Platform Leadership

Launch native mobile apps, introduce B2B white-label solutions, and pursue medical device software certification.

Phase 3 (12-18+ Months): Regional Transformation & Digital Health Leadership

- 1. Computer Vision for Medication:** Adapt the existing CV technology to identify medications and provide professionally validated guidance on how to take them, including dosage, timing, and side effects.
- 1. Predictive Health:** Develop AI-driven early warning systems for chronic diseases, a critical issue for the 650 million people in the region.
- 1. Sustainable Growth:** Build upon B2B partnerships to achieve a sustainable business model and expand into other underserved regions, a model proven in Southeast Asia.



Our Team



Jevin Tan

Jevin is a Pharmacist with 14 years of experience in the medical industries. He has recently graduated an Associate AI/ML Developer and has developed a strong skillset in cloud-based model development, including time series forecasting. For his capstone project, he created an ML platform to forecast COE prices. Jevin is passionate about applying his combined healthcare and AI expertise to solve real-world challenges and contribute to ASEAN's AI ecosystem.



Raymond Harrison

Raymond is a Singapore-based AI Engineer with a background in Design and AI from SUTD. His experience includes pioneering AI solutions at GSK's vaccines plant, developing a forgery detection app for HTX, and creating a generative model for URA. He is skilled in deep learning and PyTorch, and focuses on building software that proactively solves problems.



Jenxi Seow

Jenxi is a licensed Pharmacist with 7 years of frontline clinical practice and 10 years in China's manufacturing ecosystem, where he has guided startups and manufacturers in creating brands for global markets. As an early adopter of AI solutions, he leverages LLMs and generative AI to optimise client workflows. His firsthand experience with AI's problem-solving capabilities across Shenzhen's diverse landscape gives him unique insight into ASEAN market adoption potential.